



MOLECULAR MODELING AND DRUG DESIGNING

REPORT BY

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SUBMITTED TO: DR ISHRAT JABEEN



PROTEIN TABLE:

Pdb ID	Year of publication	Confirmational state	Organism	Resolution	Co-crystallized ligand
6VRH	2020-03-11	Outward open	Homosapien	3.3Å	Paroxetine
6VRL	2020-03-11	Outward open	Homosapien	3.8Å	I-paroxetine
7LWD	2021-08-11	Outward open	Homosapien	3.65Å	Imipramine and vilazodone
7LIA	2021-12-15	Outward open	Homosapien	3.3 Å	Serotonin
7TXT	2023-03-15	Outward open	Homosapien	3.0 Å	8090(an optimized paroxetine analogue)

Question1.

Define the activity threshold criteria?

Activity threshold criteria are **predefined cutoff values** (usually based on docking score, binding energy, or experimental bioactivity like IC₅₀, or EC₅₀) that help determine whether a ligand (compound) is likely to be biologically **active** (i.e., a good binder) or **inactive** (i.e., a weak or non-binder). The activity threshold we selected is **100nM**

- Many peer-reviewed studies use **100 nM as a cutoff** to define active ligands in **QSAR modeling, docking validation, and virtual screening**.
- A 100 nM threshold provides a **clear separation** between actives and inactive, reducing ambiguity in borderline compounds.

This threshold aligns with standard classifications:

Sr. No.	Name of Inhibitor	State	IC50 (nM)
1	Paroxetine	Active	0.08
2	Sertraline	Active	0.19
3	Duloxetine	Active	0.3162
4	Chlorpromazine	Inactive	120
5	Sulconazole	Inactive	141
6	Fluoxetine	Inactive	150

Table 1 Activity threshold criteria

Question2.

Which structure and conformation was selected, provided the PDB ID with reference. Why this structure was selected?

The structure with PDB ID: **7TXT** was selected, representing the **human serotonin transporter (SERT)** in complex with a small molecule ligand. This structure was determined using **cryo-electron microscopy (cryo-EM)** at a resolution of **3.0 Å** and its conformation is outward open with inclusion of **Allosteric Inhibitor**.

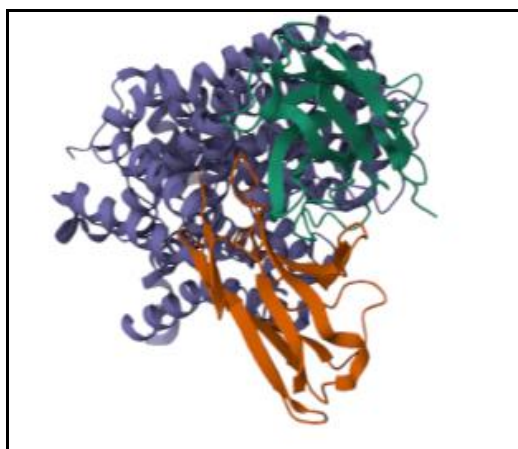


Figure 1 7TXT Protein

- **High Resolution:** 3.00 Å resolution enables accurate ligand-receptor interaction modeling.
- **Recent Publication:** Released in March 2023, reflecting the latest understanding of SERT's structure.
- **Biologically Relevant Conformation:** Captures SERT in the outward-open state, important for ligand binding.
- **Wild-Type Protein:** Represents the natural, physiological form of SERT for better docking relevance.
- **Advanced Determination Method:** Cryo-EM provides high-quality structure without crystallization, ideal for membrane proteins.

Question3.

What information does the pdbqt files contain?

PDBQT file is used in molecular docking, particularly with the AutoDock software suite. It contains several key pieces of information for both the receptor (usually a protein) and the ligand (typically a small molecule or drug candidate) to facilitate docking simulations. Here's what each part of a PDBQT file generally contains:

Field	Description
Atomic Coordinates	X, Y, Z positions of each atom
Atom Types	AutoDock-specific atom types (e.g., A, NA, OA, etc.)
Partial Charges	Assigned to each atom for electrostatic interactions
Rotatable Bonds	Defined torsional flexibility for ligands
Torsion Tree	Describes the molecular hierarchy for rotations
Torsion Root	The atom or group from which rotation is calculated

Table 2 pdbqt file

Question4.

Provide the Vina Search Space values i.e. x, y, z coordinates for Center and Dimensions. Add the names of amino acid residues in the binding cavity. Compare with the residues known from literature.

Key Binding Site Residues (TXT Protein):

- Asp98, Phe135, Phe141

Validation with Literature:

- The selected residues (**Asp98, Phe135, Phe141**) match known catalytic/active site residues important for binding from known previous studies.
- The grid box was centered to ensure all critical residues are included for accurate ligand docking.

Parameter	X	Y	Z
Center	145.7456	147.0765	136.2035
Dimensions (Å)	19.7074	16.9726	14.9236

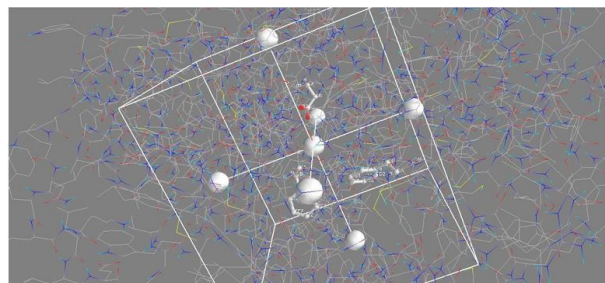


Figure 2 Grid Coordinates

Question5.

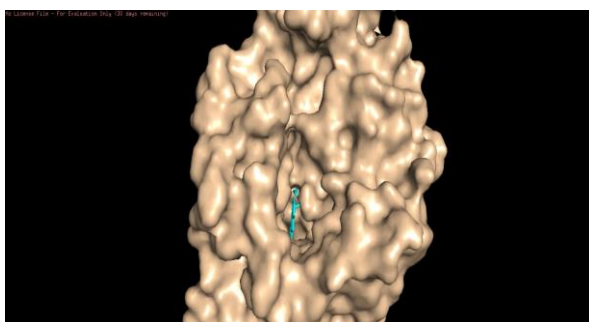
Compare all the generated models in this exercise. How many poses/ligands were generated by each participant in the group? How can you relate the binding affinity with the orientation of ligand in the binding site?

Pose Generation:

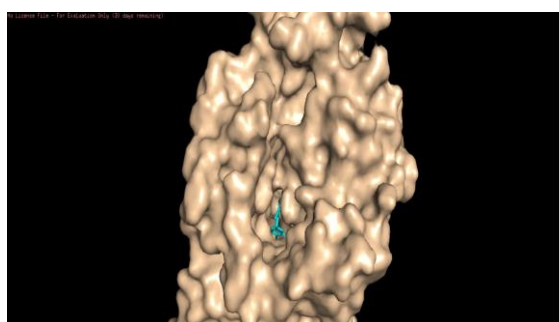
- Total ligands tested: **6** (3 active, 3 inactive)
- Total poses generated: **54** (9 poses per ligand, Vina default)

Binding Affinity vs. Ligand Orientation

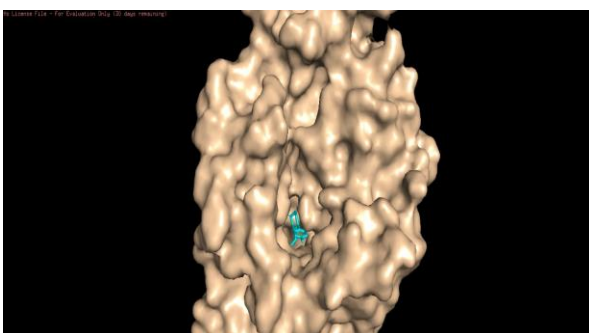
- Stronger Affinity (Lower ΔG): Poses with correct orientation into the binding pocket and optimal interactions (H-bonds, hydrophobic packing) with key residues (**e.g., Asp98, Phe135, Phe141**) typically show better (more negative) binding energies.
- Weaker Affinity: Poses with clashes, incorrect orientation, or solvent exposure tend to have less favorable ΔG .



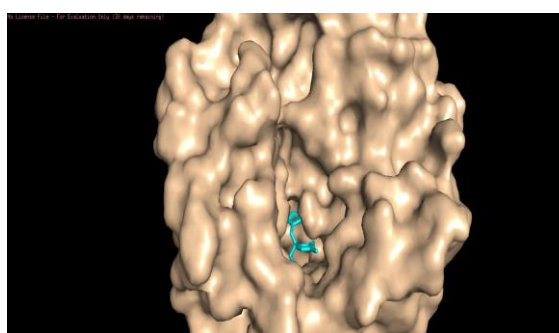
Ligand 1 CHEMBL71



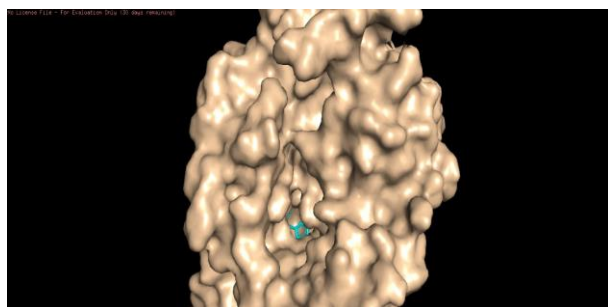
Ligand 2 CHEMBL1175



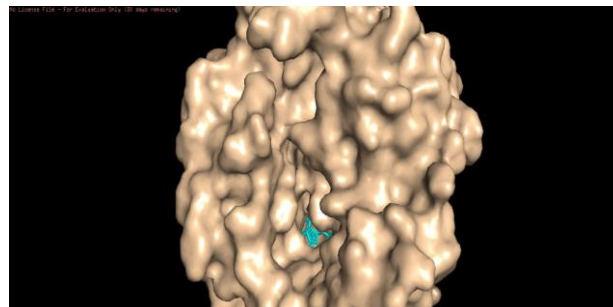
Ligand 3 CHEMBL41



Ligand 4 CHEMBL490



Ligand 5 CHEMBL809



Ligand 6 CHEMBL1221

Question6.

What are the RMSD values in AutoDock Vina? Why is the RMSD of first binding mode zero?

Root Mean Square Deviation (RMSD) measures the spatial difference between ligand poses. RMSD of first pose is always zero and it is taken as reference to calculate its interaction with other poses so its interaction with itself is zero.

1	Ligand	Binding Aff	rmsd/ub	rmsd/lb
2	7TXT_clear	-9.3	0	0
3	7TXT_clear	-9.2	7.223	4.688
4	7TXT_clear	-9	8.984	6.407
5	7TXT_clear	-9	8.943	6.42
6	7TXT_clear	-9	8.967	6.427
7	7TXT_clear	-9	8.066	5.961
8	7TXT_clear	-9	8.016	5.929
9	7TXT_clear	-9	2.861	2.073
10	7TXT_clear	-8.8	8.944	6.385
11	7TXT_clear	-7.7	0	0
12	7TXT_clear	-7.6	5.354	2.379
13	7TXT_clear	-7.5	4.998	2.405
14	7TXT_clear	-7.5	5.13	2.277
15	7TXT_clear	-7.2	5.18	2.3
16	7TXT_clear	-6.7	5.01	2.692
17	7TXT_clear	-6.6	4.688	2.272
18	7TXT_clear	-6.2	1.796	1.379
19	7TXT_clear	-6.2	5.58	2.653
20	7TXT_clear	-9.1	0	0
21	7TXT_clear	-8.7	1.585	0.723
22	7TXT_clear	-8.4	3.049	1.24
23	7TXT_clear	-8.3	7.075	3.929
24	7TXT_clear	-8.3	1.68	1.304
25	7TXT_clear	-8.1	7.025	3.805
26	7TXT_clear	-8.1	7.028	3.793

Table 3 RMSD Values

Provide the ligand-protein interactions diagram for each of the best poses within the protein.



Question8.

Can you relate the biological activity with the binding pattern of the ligands? Which ligand (in your opinion) is the most suitable inhibitor for the protein. Justify your answer?

The ligand (7TXT_clean__uff_E=323.49) shows optimal binding characteristics with a high-affinity score (-10 kcal/mol) and perfect structural alignment (0 Å RMSD), indicating stable, reproducible binding. The moderate internal energy (323.49 kcal/mol) suggests acceptable conformational strain. These computational results strongly suggest biological relevance, though experimental validation remains essential

Ligand	Binding Affinity	Rmsd/ub	Rmsd/lb.
7TXT_clean__uff_E=323.49	-10 kcal/mol	0	0

Lipophilic Efficiency (LiPE)

1. Convert IC₅₀ (nM) to Molar (M) and calculate pIC₅₀

$$pIC_{50} = -\log_{10}(IC_{50} \times 10^{-9})$$

2. Obtain logP Values

Sr.no	Name	State	IC50(nM)	IC50(M)	PIC50	logP (Predicted)
1	Paroxetine	Active	0.08	8.0×10^{-11}	10.10	3.95
2	Sertraline	Active	0.19	1.9×10^{-10}	9.72	5.20
3	Duloxetine	Active	0.3162	3.162×10^{-10}	9.50	4.25
4	Chlorpromazine	Inactive	120	1.2×10^{-7}	6.92	5.41
5	Sulconazole	Inactive	141	1.41×10^{-7}	6.85	5.80
6	Fluoxetine	Inactive	150	1.5×10^{-7}	6.82	4.05

3. Compute LiPE = pIC50 – logP

Sr. No.	Name	LiPE
1	Paroxetine	$10.10 - 3.95 = 6.15$
2	Sertraline	$9.72 - 5.20 = 4.52$
3	Duloxetine	$9.50 - 4.25 = 5.25$
4	Chlorpromazine	$6.92 - 5.41 = 1.51$
5	Sulconazole	$6.85 - 5.80 = 1.05$
6	Fluoxetine	$6.82 - 4.05 = 2.77$